Chapter 2. Basic Probability Theory

Random variables, inequalities, modes of convergence, distributions, variance, the monotone class theorem, conditional expectation

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2.1 Random variables and integration

Motivation. Probability is measure theory with total mass one: random variables, expectations, "a.s.".

$(\Omega, \mathcal{A})$ is a measurable space; $\mathbb{R}$ carries $\mathcal{B}(\mathbb{R})$, $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ carries $\mathcal{B}(\overline{\mathbb{R}}) = \{B \subset \overline{\mathbb{R}} \mid B \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\}$.

Definition 2.1 (Random variable)

A **random variable** (r.v.) on $(\Omega, \mathcal{A})$ is an $\mathcal{A}/\mathcal{B}(\overline{\mathbb{R}})$-measurable map $X : \Omega \to \overline{\mathbb{R}}$. For real-valued X it suffices to check $\mathcal{A}/\mathcal{B}(\mathbb{R})$-measurability. We write $\{X \le c\} := \{\omega \mid X(\omega) \le c\}$, $\{X < c\}$, etc.

Facts 2.2

Mostly restatements of Section 1.3:

(i) X is a r.v. iff $\{X \le c\} \in \mathcal{A}$ for all $c \in \mathbb{R}$ ($\sigma(\{[-\infty, c]\}) = \mathcal{B}(\overline{\mathbb{R}})$), and c in a countable dense S (e.g. $\mathbb{Q}$) suffices since $\{X > c\} = \bigcup_{q \in S,\, q > c} \{X > q\}$.

(ii) If $\mathcal{A} = \mathcal{P}(\Omega)$, every function $\Omega \to \overline{\mathbb{R}}$ is a r.v.

(iii) If X is a r.v. and $h : \mathbb{R} \to \mathbb{R}$ (or $\overline{\mathbb{R}} \to \overline{\mathbb{R}}$) is Borel measurable, then $h(X)$ is a r.v. (preimages compose); e.g. $|X|$, X^2, $|X|^p$, e^X.

(iv) For r.v.'s $X_1, X_2, \ldots$ and $\alpha_i \in \mathbb{R}$, the following are r.v.'s: $\sum_{i \le n} \alpha_i X_i$ whenever no $\infty - \infty$ occurs; $\sum_{i=1}^{\infty} \alpha_i X_i$ if all $\alpha_i, X_i \ge 0$; $\sup_i X_i$, $\inf_i X_i$, $X_1 \wedge X_2 := \min$, $X_1 \vee X_2 := \max$; $\limsup_i X_i$, $\liminf_i X_i$, and $\lim_i X_i$ if it exists.

(v) $\mathbb{1}_A$ is a r.v. for $A \in \mathcal{A}$. An **elementary** (simple) r.v. is $X = \sum_{i=1}^{n} c_i \mathbb{1}_{A_i}$ with $c_i \in \mathbb{R}$ and disjoint $A_i \in \mathcal{A}$. Every r.v. is a limit of elementary ones (Theorem 1.40; $X = X^+ - X^-$, $X^+ := X \vee 0$, $X^- := -(X \wedge 0)$); for $X \ge 0$,

$$X_n := \sum_{i=0}^{n2^n - 1} \frac{i}{2^n} \mathbb{1}_{\{\frac{i}{2^n} \le X < \frac{i+1}{2^n}\}} + n \mathbb{1}_{\{X \ge n\}} \nearrow X.$$

On $(\Omega, \mathcal{A}, P)$: $\int \mathbb{1}_A \, dP = P(A)$; $\int \sum c_i \mathbb{1}_{A_i} \, dP = \sum c_i P(A_i)$; $\int X \, dP = \lim_n \int X_n \, dP$ for elementary $X_n \nearrow X \ge 0$ (MCT); and if $\min(\int X^+ dP, \int X^- dP) < \infty$,

$$\mathbb{E}[X] := \int_\Omega X \, dP := \int X^+ dP - \int X^- dP \in \overline{\mathbb{R}}$$

is the **expectation** (also $\int X(\omega) \, P(d\omega)$). Up to Theorem 2.8 everything translates Section 1.4.

Definition 2.3 ($\mathcal{L}^p$ and almost sure)

(i) $\mathcal{L}^1 := \mathcal{L}^1(\Omega, \mathcal{A}, P) := \{X : \Omega \to \mathbb{R} \text{ r.v.} \mid \mathbb{E}[|X|] < \infty\}$, and for $p \in [1, \infty)$, $\mathcal{L}^p := \{X \text{ real r.v.} \mid \mathbb{E}[|X|^p] < \infty\}$.

(ii) A property holds **P-almost surely** (P-a.s.) if it holds outside some $N \in \mathcal{A}$ with $P(N) = 0$.

With $\mathcal{N}_0 := \{X \mid X = 0 \ P\text{-a.s.}\}$, $L^p := \mathcal{L}^p / \mathcal{N}_0$ is a Banach space for $\|X\|_p := \mathbb{E}[|X|^p]^{1/p}$ (Theorem 1.70). The expectation is monotone ($0 \le X \le Y$ a.s. $\Rightarrow 0 \le \mathbb{E}[X] \le \mathbb{E}[Y]$) and linear on $\mathcal{L}^1$ (Theorem 1.54).

2.2 Inequalities

Motivation. Jensen, Markov–Chebyshev and Hölder–Minkowski from Section 1.5, for $P(\Omega) = 1$.

Proposition 2.4 (Jensen's inequality)

Let φ be convex on an open interval $I \subset \mathbb{R}$ and $X \in \mathcal{L}^1$ with $X(\Omega) \subset I$. Then $\mathbb{E}[X] \in I$, $(\varphi \circ X)^- \in \mathcal{L}^1$ and $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.

Proof. Theorem 1.63 on $(\Omega, \mathcal{A}, P)$ (verbatim for infinite endpoints). ■

Example 2.5

For $X \in \mathcal{L}^1$: $\mathbb{E}[X]^2 \leq \mathbb{E}[X^2]$ ($\varphi(x) = x^2$). Moreover, for $1 \leq p \leq q$ and any r.v. X (integrability not needed),

$$\|X\|_p = \mathbb{E}[|X|^p]^{1/p} \leq \mathbb{E}[|X|^q]^{1/q} = \|X\|_q.$$

Proof. Jensen with convex $|x|^{q/p}$ on the bounded $(|X| \wedge n)^p$: $\mathbb{E}[(|X| \wedge n)^p]^{q/p} \leq \mathbb{E}[(|X| \wedge n)^q]$; take the $1/q$ power, $n \to \infty$ (MCT).

Corollary 2.6

Let X be a r.v.

(i) (Markov) $P(|X| \geq c) \leq \frac{1}{c}\mathbb{E}[|X|]$ for $c > 0$. In particular $\mathbb{E}[|X|] = 0 \Rightarrow X = 0$ a.s., and $\mathbb{E}[|X|] < \infty \Rightarrow |X| < \infty$ a.s.

(ii) (Chebyshev) For $X \in \mathcal{L}^1$, $P(|X - \mathbb{E}[X]| \geq c) \leq \frac{\mathrm{Var}(X)}{c^2}$, where $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2]$ is the **variance**. More generally $P(|X| \geq c) \leq c^{-p}\mathbb{E}[|X|^p]$ for $p > 0$.

Proof. Proposition 1.61 for X, resp. $X - \mathbb{E}[X]$; then Lemma 1.62(i) and Theorem 1.52(iii). ■

Theorem 2.7

(i) (Hölder) For $p, q \in (1, \infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $X \in \mathcal{L}^p$, $Y \in \mathcal{L}^q$: $\|XY\|_1 \leq \|X\|_p \|Y\|_q$. For $p = q = 2$ this is the Cauchy–Schwarz inequality in the Hilbert space L^2. (ii) (Minkowski) For $p \geq 1$ and $X, Y \in \mathcal{L}^p$: $\|X + Y\|_p \leq \|X\|_p + \|Y\|_p$.

Proof. Theorem 1.67(ii) and Step 1 of the proof of Theorem 1.67(i). ■

2.3 Convergence of random variables

Motivation. Almost sure, in probability, and L^p convergence differ; Borel–Cantelli links them.

Theorem 2.8 (Limit theorems)

(i) (MCT) $0 \leq X_1 \leq X_2 \leq \ldots \Rightarrow \lim_n \mathbb{E}[X_n] = \mathbb{E}[\lim_n X_n]$.

(ii) (Fatou) $X_n \geq 0 \Rightarrow \mathbb{E}[\liminf_n X_n] \leq \liminf_n \mathbb{E}[X_n]$.

(iii) (DCT) If $|X_n| \leq Y \in \mathcal{L}^1$ a.s. for all n and $\lim_n X_n$ exists a.s., then $\mathbb{E}[\lim_n X_n] =$

$\lim_n \mathbb{E}[X_n]$. Here $\lim_n X_n$ means any r.v. X with $X = \lim_n X_n$ a.s., e.g. $X = \lim_n X_n \mathbb{1}_{\Omega\setminus N}$ with $P(N) = 0$.

Proof. Theorems 1.53, 1.57, 1.58 together with Remark 1.59; the a.s. limit $\lim_n X_n \mathbb{1}_{\Omega\setminus N}$ is measurable by Corollary 1.36(ii). ■

Events. For $A_n \in \mathcal{A}$: $\bigcup_i A_i$ = "at least one A_i occurs", $\bigcap_i A_i$ = "all occur",

$$\limsup_n A_n := \bigcap_n \bigcup_{m\geq n} A_m = \text{"infinitely many } A_n \text{ occur" } (A_n \text{ i.o.}),$$
$$\liminf_n A_n := \bigcup_n \bigcap_{m\geq n} A_m = \text{"all but finitely many } A_n \text{ occur"},$$

and $\liminf_n A_n \subset \limsup_n A_n$.

Lemma 2.9 (Borel–Cantelli, part 1)

$\sum_n P(A_n) < \infty \;\Rightarrow\; P(\limsup_n A_n) = 0$.

Proof. $\bigcup_{m\geq n} A_m \searrow \limsup_n A_n$, so by continuity from above and subadditivity $P(\limsup_n A_n) = \lim_n P(\bigcup_{m\geq n} A_m) \leq \lim_n \sum_{m\geq n} P(A_m) = 0$. ■

Lemma 2.10 (Fast convergence in probability)

If $\sum_n P(|X_n| \geq \varepsilon) < \infty$ for every $\varepsilon > 0$, then $P(\lim_n X_n = 0) = 1$.

Proof.

Step 1. For $\varepsilon > 0$ let $N_\varepsilon := \limsup_n \{|X_n| \geq \varepsilon\}$; Borel–Cantelli gives $P(N_\varepsilon) = 0$. For $\omega \notin N_{1/k}$,

$$|X_m(\omega)| < \tfrac{1}{k} \quad \text{for all but finitely many } m. \tag{2.1}$$

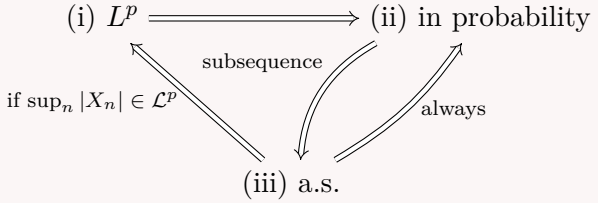
Step 2. $N := \bigcup_k N_{1/k}$ is null, and for $\omega \notin N$, (2.1) holds for every k, i.e. $X_n(\omega) \to 0$. ■

Definition 2.11 (Modes of convergence)

For real r.v.'s $X, X_1, X_2, \ldots$: (i) **L^p-convergence** ($p \geq 1$): $\|X_n - X\|_p \to 0$; (ii) **convergence in probability**: $P(|X_n - X| \geq \varepsilon) \to 0$ for every $\varepsilon > 0$; (iii) **P-a.s. convergence**: $P(\lim_n X_n = X) = 1$.

Theorem 2.12 (Comparison of the modes)

(i)⇒(ii); (iii)⇒(ii); (ii)⇒(iii) along a subsequence; (iii)⇒(i) if $\sup_n |X_n| \in \mathcal{L}^p$.



Proof.

Step (i)⇒(ii). Chebyshev: $P(|X_n - X| \geq \varepsilon) \leq \varepsilon^{-p}\mathbb{E}[|X_n - X|^p]$.

Step (iii)⇒(ii). $\{\lim_n X_n = X\} = \bigcap_k A_k$ with $A_k := \bigcup_m \bigcap_{n\ge m}\{|X_n - X| \le \frac{1}{k}\}$. So $P(A_k) = 1$, and continuity from below gives

$$1 = P(A_k) = \lim_m P\Big(\bigcap_{n\ge m}\{|X_n - X| \le \tfrac{1}{k}\}\Big) \le \liminf_m P(|X_m - X| \le \tfrac{1}{k}) \le 1.$$

Hence $P(|X_m - X| > \frac{1}{k}) \to 0$ for every k.

Step (iii)⇒(i). With $Y := \sup_n |X_n|$ and $|X| \le Y$ a.s., $|X_n - X|^p \le 2^p Y^p \in \mathcal{L}^1$; apply the DCT.

Step (ii)⇒(iii) along a subsequence. Take $P(|X_{n_k} - X| \ge \frac{1}{k}) < 2^{-k}$; for $\frac{1}{k} < \varepsilon$, $P(|X_{n_k} - X| \ge \varepsilon) \le 2^{-k}$, so $\sum_k P(|X_{n_k} - X| \ge \varepsilon) < \infty$ and Lemma 2.10 applies. ∎

2.4 Image measure and distribution of random variables

Motivation. A random variable matters through its distribution P_X (or F_X) on $\mathbb{R}$.

$(\Omega, \mathcal{A})$, $(\tilde\Omega, \tilde{\mathcal{A}})$ are measurable spaces.

Proposition 2.13 (Image measure)

If $T : \Omega \to \tilde\Omega$ is $\mathcal{A}/\tilde{\mathcal{A}}$-measurable and P a probability measure on $(\Omega, \mathcal{A})$, then $\tilde P(\tilde A) := P(T \in \tilde A) = P(T^{-1}(\tilde A))$, $\tilde A \in \tilde{\mathcal{A}}$, is a probability measure on $(\tilde\Omega, \tilde{\mathcal{A}})$, the **image** (push-forward, induced) measure, written $P \circ T^{-1}$, $T(P)$ or P_T.

Proof. $\tilde P(\emptyset) = 0$, $\tilde P(\tilde\Omega) = 1$; preimages of disjoint sets are disjoint and commute with unions, so $\tilde P(\bigcup_i \tilde A_i) = \sum_i \tilde P(\tilde A_i)$. ∎

Definition 2.14 (Distribution, distribution function)

The **distribution** of a real r.v. X on $(\Omega, \mathcal{A}, P)$ is $P_X(B) := P(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$. Its **distribution function** is

$$F_X(t) := P(X \le t) = P_X((-\infty, t]), \qquad t \in \mathbb{R}. \tag{2.2}$$

For μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $F(x) := \mu((-\infty, x])$. (Extended real X: distribution on $\mathcal{B}(\overline{\mathbb{R}})$, but no distribution function.)

Remark 2.15

(i) F is increasing, right continuous, $F(-\infty) = 0$, $F(+\infty) = 1$, by continuity of μ: $(-\infty, b_n] \searrow (-\infty, a]$ for $b_n \searrow a$, $(-\infty, a] \searrow \emptyset$, $(-\infty, b] \nearrow \mathbb{R}$.

Conversely each such F is the distribution function of a unique μ on $\mathcal{B}(\mathbb{R})$, the distribution of $X(x) = x$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$.

[Supplement] *Proof.* Uniqueness: $\{(-\infty, t]\}$ is a $\cap$-stable generator (Theorem 1.9). Existence: on $((0,1), \mathcal{B}, \lambda)$, $G(u) := \inf\{t \mid F(t) \ge u\} \in \mathbb{R}$ attains its infimum (right continuity), so $G(u) \le t \iff u \le F(t)$, $\lambda(G \le t) = F(t)$, and $\mu := \lambda \circ G^{-1}$.

(ii) For Borel $\Phi : \mathbb{R} \to \mathbb{R}$ with $\Phi(X) \in \mathcal{L}^1$ or $\Phi \ge 0$,

$$\mathbb{E}[\Phi(X)] = \int_\Omega \Phi(X)\, dP = \int_{\mathbb{R}} \Phi\, dP_X = \int_{\mathbb{R}} \Phi(s)\, dF_X(s) = \int_{\mathbb{R}} \Phi(s) F_X'(s)\, ds.$$

The third equality is notation; the last needs $F_X \in C^1$ (or absolutely continuous).

[Supplement] *Second equality (transformation theorem).* Indicators, simple $\Phi \geq 0$, MCT, then $\Phi^+ - \Phi^-$.

Last equality. $F_X(x) - F_X(-n) = \int_{-n}^{x} F_X' \, dt$ (FTC, Theorem 1.60); $n \to \infty$ (MCT) gives density F_X' (Remark 2.18(ii)); then repeat the indicator–simple–MCT argument.

Remark 2.16

(i) The **jump** $F(x) - F(x-) = \lim_n \mu((x - \frac{1}{n}, x]) = \mu(\{x\})$; F is continuous iff μ has no atoms ("μ is continuous").

(ii) An increasing bounded F has at most countably many discontinuities. [Supplement] *Reason:* at most k jumps are $\geq 1/k$.

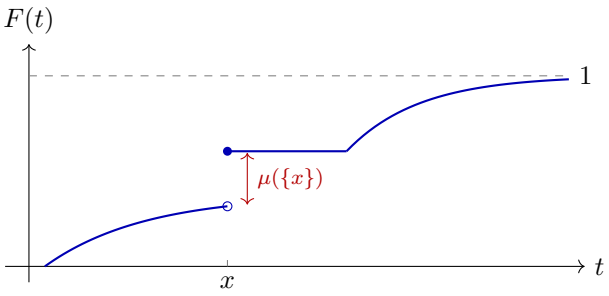


Figure. *A distribution function; its jump at x is the atom $\mu(\{x\})$.*

Definition 2.17 (Discrete and absolutely continuous distributions)

(i) F (or μ) is **discrete** if $\mu(S) = 1$ for a countable $S \subset \mathbb{R}$; then $F(x) = \sum_{y \in S,\, y \leq x} \mu(\{y\})$ is a step function.

(ii) F (or μ) is **absolutely continuous** if there is a measurable **density** $f \geq 0$ with

$$F(x) = \int_{-\infty}^{x} f(t) \, dt, \tag{2.3}$$

respectively

$$\mu(A) = \int_A f(t) \, dt = \int_{-\infty}^{\infty} \mathbb{1}_A f \, dt \quad \text{for all } A \in \mathcal{B}(\mathbb{R}). \tag{2.4}$$

In particular $\int_{-\infty}^{\infty} f \, dt = 1$.

Remark 2.18

(i) Every measurable $f \geq 0$ with $\int f \, dt = 1$ defines a probability measure $A \mapsto \int_A f \, dt$ (Theorem 1.47). (ii) (2.3)⇒(2.4) by uniqueness on the generator $\{(-\infty, x]\}$ (Theorem 1.9); the converse is trivial.

Example 2.19

(i) (*Uniform on $[a,b]$*) $f = \frac{1}{b-a} \mathbb{1}_{[a,b]}$, $F(x) = 0$ for $x \leq a$, $\frac{x-a}{b-a}$ on $[a,b]$, 1 for $x \geq b$.

(ii) (*Exponential with parameter $\alpha > 0$*) $f(x) = \alpha e^{-\alpha x} \mathbb{1}_{[0,\infty)}(x)$, $F(x) = (1 - e^{-\alpha x}) \mathbb{1}_{[0,\infty)}(x)$. Continuous geometric analogue: $\int_k^{k+1} f = e^{-\alpha k}(1 - e^{-\alpha}) = (1-p)^k p$ with $p := 1 - e^{-\alpha}$.

(iii) (*Normal* $N(m,\sigma^2)$) Density $f_{m,\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-m)^2/(2\sigma^2)}$. $z = (y-m)/\sigma$ gives $F_{m,\sigma^2}(x) = F_{0,1}(\frac{x-m}{\sigma})$ (no closed form), $\mathbb{E}[X] = m$, $\mathrm{Var}(X) = \sigma^2$. [Supplement] *Reason:* $\mathbb{E}[X] - m = \sigma \int z f_{0,1}\, dz = 0$ (symmetry), $\mathrm{Var}(X) = \sigma^2 \int z^2 f_{0,1}\, dz = \sigma^2$ (parts: $\int z \cdot z e^{-z^2/2} dz = \int e^{-z^2/2} dz$). Central moments:

$$\mathbb{E}[|X-m|^p] = 2\int_0^\infty x^p \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}\, dx = \frac{2^{p/2}\sigma^p}{\sqrt{\pi}}\,\Gamma\Big(\frac{p+1}{2}\Big) \qquad (y = x^2/(2\sigma^2)),$$

(iv) (*Binomial*) $\Omega = S^n$ uniform, success $x_i \in S_0 \subset S$, $p := |S_0|/|S|$; "exactly k successes" has $P(A_k) = \binom{n}{k}|S_0|^k|S\setminus S_0|^{n-k}/|S|^n = \binom{n}{k}p^k(1-p)^{n-k}$. The **binomial distribution** is $B(n,p) := \sum_{k=0}^n \binom{n}{k}p^k(1-p)^{n-k}\delta_k$.

(v) (*Poisson limit*) With $p_n = \lambda/n$,

$$\binom{n}{k}\Big(\frac{\lambda}{n}\Big)^k\Big(1-\frac{\lambda}{n}\Big)^{n-k} = \frac{\lambda^k}{k!}\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1}\underbrace{\Big(1-\frac{\lambda}{n}\Big)^{n-k}}_{\to e^{-\lambda}} \to \frac{\lambda^k}{k!}e^{-\lambda}.$$

so $B(n,p) \approx$ the **Poisson distribution** $\pi_\lambda := e^{-\lambda}\sum_{j\ge 0}\frac{\lambda^j}{j!}\delta_j$ with $\lambda = np$.

2.5 Variance and covariance

Motivation. Variance is the risk of the best constant prediction; uncorrelatedness makes variances add.

Example 2.20

For the uniform distribution on $\{\omega_1,\dots,\omega_n\}$, $\mathbb{E}[X] = \frac{1}{n}\sum_i X(\omega_i)$ is the arithmetic mean.

Definition 2.21 (Variance)

For $X \in \mathcal{L}^1$, $\mathrm{Var}(X) := \sigma^2(X) := \mathbb{E}[(X-\mathbb{E}[X])^2] \in [0,\infty]$ (the mean square prediction error).

Remark 2.22

For $X \in \mathcal{L}^1$: (i) $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; (ii) $\mathrm{Var}(X) = 0 \iff P(X = \mathbb{E}[X]) = 1$; (iii) $\mathrm{Var}(X) < \infty \iff X \in \mathcal{L}^2$.
[Supplement] *Proof.*
(i) Expand $(X-m)^2$, $m = \mathbb{E}[X]$ (both sides ∞ if $\mathbb{E}[X^2] = \infty$).
(ii) Corollary 2.6(i) applied to $(X-m)^2$.
(iii) Use (i) and $X^2 \le 2(X-m)^2 + 2m^2$.

Definition 2.23 (Covariance)

For $X, Y \in \mathcal{L}^2$ (so $XY \in \mathcal{L}^1$ by Cauchy–Schwarz)

$$\mathrm{Cov}(X,Y) := \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{2.5}$$

Remark 2.24

(i) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$ for $X \in \mathcal{L}^1$. (ii) $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$ for $X, Y \in \mathcal{L}^2$. Both follow by expanding the square, using linearity.

Definition 2.25 (Uncorrelated)

$X, Y \in \mathcal{L}^2$ are **uncorrelated** if

$$\operatorname{Cov}(X,Y) = 0 \quad (\iff \operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)). \tag{2.6}$$

Remark 2.26 (Bienaymé)

If $X_1, \dots, X_n \in \mathcal{L}^2$ are pairwise uncorrelated, then $\operatorname{Var}\left(\sum_i X_i\right) = \sum_i \operatorname{Var}(X_i)$, by expanding $\operatorname{Var}(\sum_i X_i) = \sum_{i,j} \operatorname{Cov}(X_i, X_j)$.

2.6 The monotone class theorem

Motivation. An identity on a multiplicative class that survives monotone limits holds for all bounded measurable functions.

Definition 2.27 (Monotone vector space)

A vector space $\mathcal{H}$ of bounded real functions on Ω is a **monotone vector space** (MVS) if (i) $1 = \mathbb{1}_\Omega \in \mathcal{H}$, and (ii) whenever $f_n \in \mathcal{H}$ with $0 \le f_1 \le f_2 \le \cdots \nearrow f$ and f bounded, then $f \in \mathcal{H}$.

Lemma 2.28

An MVS is closed under uniform convergence.

Proof. Let $f_n \in \mathcal{H}$ with $\|f - f_n\|_\infty := \sup_\omega |f(\omega) - f_n(\omega)| \to 0$.

Step 1. Assume $f \ge 0$ (add the constant $\|f\|_\infty \in \mathcal{H}$).

Step 2. Take summable $\varepsilon_k := \|f_{n_k} - f_{n_{k+1}}\|_\infty$, $a_k := \sum_{n \ge k} \varepsilon_n \searrow 0$, $g_k := f_{n_k} - a_k + 2a_1 \in \mathcal{H}$: $g_{k+1} - g_k = f_{n_{k+1}} - f_{n_k} + \varepsilon_k \ge 0$, $g_k \nearrow f + 2a_1$ (bounded), and $g_1 = f_{n_1} + a_1 \ge f \ge 0$ because $f - f_{n_1} = \sum_k (f_{n_{k+1}} - f_{n_k}) \le a_1$.

Step 3. By Definition 2.27(ii), $f + 2a_1 \in \mathcal{H}$, hence $f \in \mathcal{H}$. ∎

Notation. For a class $\mathcal{M}$ of functions on Ω, $\sigma(\mathcal{M})$ is the smallest σ-algebra making all $f \in \mathcal{M}$ measurable, and $\sigma(\mathcal{M})_b$ the bounded $\sigma(\mathcal{M})$-measurable functions.

Theorem 2.29 (Monotone class theorem, algebra form)

Let $\mathcal{M}$ be a multiplicative class of bounded functions ($f, g \in \mathcal{M} \Rightarrow fg \in \mathcal{M}$) and $\mathcal{H}$ an MVS with $\mathcal{M} \subset \mathcal{H}$. Then $\sigma(\mathcal{M})_b \subset \mathcal{H}$.

Proof.

Step 1 (algebra). $\mathcal{M}_0 := \operatorname{span}(\mathcal{M}, 1) \subset \mathcal{H}$ is an algebra; its uniform closure $\overline{\mathcal{M}_0}$ is an algebra in $\mathcal{H}$ (Lemma 2.28) with $\sigma(\overline{\mathcal{M}_0}) = \sigma(\mathcal{M})$.

Step 2 (truncation). $f \in \overline{\mathcal{M}_0} \Rightarrow f \wedge \alpha \in \overline{\mathcal{M}_0}$: Weierstrass polynomials P_ε with $|x \wedge \alpha - P_\varepsilon(x)| < \varepsilon$ on $[-\|f\|_\infty, \|f\|_\infty]$ give $\|f \wedge \alpha - P_\varepsilon(f)\|_\infty < \varepsilon$.

Step 3 (reduction to indicators). $f \in \sigma(\mathcal{M})_b$ is a uniform limit of elementary functions (Note 1.41), so it suffices that $\mathcal{S} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \mathbb{1}_A \in \mathcal{H}\} = \sigma(\overline{\mathcal{M}_0})$.

Step 4 ($\mathcal{S}$ is a Dynkin system). $\Omega \in \mathcal{S}$ since $1 \in \mathcal{H}$; $\mathbb{1}_{A^c} = 1 - \mathbb{1}_A$; for disjoint A_i, $\sum_{i \le n} \mathbb{1}_{A_i} \nearrow \mathbb{1}_{\bigcup A_i}$, so MVS (ii) applies.

Step 5 (a $\cap$-stable generator). Let $\mathcal{E} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \exists f_n \in \overline{\mathcal{M}_0},\ f_n \ge 0,\ f_n \nearrow \mathbb{1}_A\}$. $\mathcal{E}$ is $\cap$-stable (if $f_n \nearrow \mathbb{1}_A$, $g_n \nearrow \mathbb{1}_B$, then $f_n g_n \nearrow \mathbb{1}_{A \cap B}$), and $\mathcal{E} \subset \mathcal{S}$ by MVS (ii). Theorem 1.8 gives

$$\sigma(\mathcal{E}) = \mathcal{D}(\mathcal{E}) \subset \mathcal{S} \subset \sigma(\overline{\mathcal{M}_0}). \tag{2.7}$$

Step 6. $(n(f-\alpha)^+) \wedge 1 \in \overline{\mathcal{M}_0}$ (Step 2; $(f-\alpha)^+ = (f-\alpha) - ((f-\alpha) \wedge 0)$) increases to $\mathbb{1}_{\{f > \alpha\}}$, so the generating sets $\{f > \alpha\}$ lie in $\mathcal{E}$ and (2.7) is an equality. ∎

Example 2.30

Let μ_1, μ_2 be probability measures on a metric space (S, d) with $\int f\, d\mu_1 = \int f\, d\mu_2$ for all $f \in C_b(S)$ (bounded continuous). Then $\mu_1 = \mu_2$.
Proof. $\mathcal{H} := \{f \in \mathcal{B}(S)_b \mid \int f\, d\mu_1 = \int f\, d\mu_2\}$ is an MVS (MCT) containing $C_b(S)$, so $\mathcal{H} \supset \sigma(C_b(S))_b$. Closed $F = \{d(\cdot, F) \wedge 1 = 0\} \in \sigma(C_b(S))$, so $\sigma(C_b(S)) = \mathcal{B}(S)$ and $\mu_1 = \mu_2$.

2.7 Conditional expectation

Motivation. The best guess given $\mathcal{A}$: the L^2 projection, characterised by $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$, which extends to $\mathcal{L}^1$.

$(\Omega, \mathcal{F}, P)$ is a probability space, $\mathcal{A} \subset \mathcal{F}$ a sub-σ-algebra.

Heuristics. For $X \in L^2(\Omega, \mathcal{F}, P)$ seek $Y \in L^2(\Omega, \mathcal{A}, P)$ with

$$\mathbb{E}[(X-Y)^2] \le \mathbb{E}[(X-Y_0)^2] \qquad \forall Y_0 \in L^2(\Omega, \mathcal{A}, P). \tag{2.8}$$

Then for $Z \in L^2(\mathcal{A})$, $\varepsilon \neq 0$, $\mathbb{E}[(X-Y)^2] \le \mathbb{E}[(X-Y-\varepsilon Z)^2]$ gives $2\varepsilon\mathbb{E}[(X-Y)Z] \le \varepsilon^2\mathbb{E}[Z^2]$, and $\varepsilon \to 0^\pm$ gives $\mathbb{E}[XZ] = \mathbb{E}[YZ]$. Y is the orthogonal projection onto the closed $L^2(\Omega, \mathcal{A}, P)$.

Definition 2.31 (Conditional expectation)

For $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$, a **conditional expectation of X given $\mathcal{A}$** is any r.v. Y with (i) Y is $\mathcal{A}$-measurable ("$Y \in \mathcal{A}$"), and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is integrable and is a.s. unique; we write $Y = \mathbb{E}[X \mid \mathcal{A}]$.

Integrability. With $A = \{Y > 0\} \in \mathcal{A}$: $\mathbb{E}[Y^+] = \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A] \le \mathbb{E}[|X|]$ and $\mathbb{E}[Y^-] = \mathbb{E}[-X\mathbb{1}_{A^c}] \le \mathbb{E}[|X|]$.

Uniqueness. For another Y', $A_n := \{Y - Y' \ge \frac{1}{n}\} \in \mathcal{A}$ gives $0 = \mathbb{E}[(Y-Y')\mathbb{1}_{A_n}] \ge \frac{1}{n}P(A_n)$, so $P(Y > Y') \le \sum_n P(A_n) = 0$; symmetrically $P(Y' > Y) = 0$.

Existence follows from Radon–Nikodym ($\mathbb{E}[X \mid \mathcal{A}]$ as a density).

Definition 2.32 (Absolute continuity)

For finite measures ν, μ on $(\Omega, \mathcal{A})$, ν is **absolutely continuous** w.r.t. μ, $\nu \ll \mu$, if $\mu(N) = 0 \Rightarrow \nu(N) = 0$ for $N \in \mathcal{A}$.

Theorem 2.33 (Radon–Nikodym)

Let ν, μ be finite measures on $(\Omega, \mathcal{A})$ with $\nu \ll \mu$. Then there is $\varphi \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, $\varphi \geq 0$, with

$$\nu(A) = \int_A \varphi \, d\mu \qquad \text{for all } A \in \mathcal{A},$$

and φ is μ-a.e. unique. It is the **Radon–Nikodym derivative** $\frac{d\nu}{d\mu}$. (The theorem extends to σ-finite ν, μ, without integrability of the density.)

Proof [Supplement] (von Neumann).

Step 1 (a representing function). $\rho := \mu + \nu$; $|\int h \, d\nu| \leq \int |h| \, d\rho \leq \rho(\Omega)^{1/2} \|h\|_{L^2(\rho)}$, so by Riesz on the Hilbert space $L^2(\rho)$ (Theorem 1.70) some $g \in L^2(\rho)$ has

$$\int h \, d\nu = \int hg \, d\rho \qquad \forall h \in L^2(\rho). \tag{2.9}$$

Step 2 ($0 \leq g \leq 1$). With $h = \mathbb{1}_{\{g<0\}}$: $0 \leq \nu(g < 0) = \int_{\{g<0\}} g \, d\rho \leq 0$, so $\rho(g < 0) = 0$. With $h = \mathbb{1}_{\{g>1\}}$: $\nu(g > 1) = \int_{\{g>1\}} g \, d\rho \geq \rho(g > 1) \geq \nu(g > 1)$, so $\rho(g > 1) = 0$; modify g on a null set to get $0 \leq g \leq 1$.

Step 3. $\int h(1-g) \, d\nu = \int hg \, d\mu$ for bounded, hence (MCT) all $h \geq 0$. $N := \{g = 1\}$: $\mu(N) = \int_N (1-g) \, d\nu = 0$, so $\nu(N) = 0$.

Step 4. $h = \mathbb{1}_A(1 + g + \cdots + g^n)$: $\int_A (1 - g^{n+1}) \, d\nu = \int_A g(1 + \cdots + g^n) \, d\mu$; with $1 - g^{n+1} \nearrow \mathbb{1}_{N^c}$, $g(1 + \cdots + g^n) \nearrow \varphi := \frac{g}{1-g} \mathbb{1}_{N^c}$ and the MCT, $\nu(A) = \nu(A \cap N^c) = \int_A \varphi \, d\mu$; $A = \Omega$ gives $\varphi \in \mathcal{L}^1$.

Step 5 (uniqueness). $\int_A (\varphi - \varphi') \, d\mu = 0$ for all A; Lemma 1.62(ii). ∎

Existence of $\mathbb{E}[X \mid \mathcal{A}]$. $\nu^{\pm}(A) := \int_A X^{\pm} \, dP$ on $\mathcal{A}$ are finite, $\nu^{\pm} \ll P|_{\mathcal{A}}$; with $Y := \frac{d\nu^+}{dP|_{\mathcal{A}}} - \frac{d\nu^-}{dP|_{\mathcal{A}}}$ (Theorem 2.33), for $A \in \mathcal{A}$

$$\int_A X \, dP = \nu^+(A) - \nu^-(A) = \int_A \frac{d\nu^+}{dP|_{\mathcal{A}}} \, dP - \int_A \frac{d\nu^-}{dP|_{\mathcal{A}}} \, dP = \int_A Y \, dP,$$

so $Y = \mathbb{E}[X \mid \mathcal{A}]$. With $A = \Omega$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$.

Interpretation. $\mathbb{E}[X \mid \mathcal{A}]$ is the best guess for X knowing which $A \in \mathcal{A}$ occurred.

Example 2.34

Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$.

(i) X $\mathcal{A}$-measurable $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$.

(ii) $\mathcal{A} = \{\emptyset, \Omega\} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$.

(iii) $\mathcal{A} = \{\emptyset, A, A^c, \Omega\}$ with $0 < P(A) < 1 \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \frac{\mathbb{E}[X\mathbb{1}_A]}{P(A)} \mathbb{1}_A + \frac{\mathbb{E}[X\mathbb{1}_{A^c}]}{P(A^c)} \mathbb{1}_{A^c}$.

(iv) If X is independent of $\mathcal{A}$ (i.e. $\mathbb{1}_A$ and X are independent for all $A \in \mathcal{A}$, Section 3.4), then $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\mathbb{E}[\mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X]\mathbb{1}_A]$, so $\mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$.

Check Definition 2.31 directly.

Theorem 2.35 (Properties of conditional expectation)

Let $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$; (in)equalities hold P-a.s.

(i) (Linearity) $\mathbb{E}[\alpha X + \beta Y \mid \mathcal{A}] = \alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}]$.

(ii) (Monotonicity) $X \le Y \Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

(iii) (Pulling out what is known) Y $\mathcal{A}$-measurable, $XY \in \mathcal{L}^1 \Rightarrow \mathbb{E}[XY \mid \mathcal{A}] = Y\mathbb{E}[X \mid \mathcal{A}]$.

(iv) (Tower property) For $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{F}$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_1] \mid \mathcal{A}_2] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1] = \mathbb{E}[X \mid \mathcal{A}_1]$.

(v) (Contraction) $|\mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[|X| \mid \mathcal{A}]$.

(vi) (DCT) If $|X_n| \le Y \in \mathcal{L}^1$ and $X_n \to X$ a.s., then $\mathbb{E}[X_n \mid \mathcal{A}] \to \mathbb{E}[X \mid \mathcal{A}]$ in $L^1(\Omega, \mathcal{A}, P)$ and a.s.

(vii) (Jensen) $\varphi : \mathbb{R} \to \mathbb{R}$ convex, $\varphi(X) \in \mathcal{L}^1 \Rightarrow \varphi(\mathbb{E}[X \mid \mathcal{A}]) \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$.

Proof. (i) For $A \in \mathcal{A}$, $\mathbb{E}[(\alpha X + \beta Y)\mathbb{1}_A] = \alpha\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]\mathbb{1}_A] + \beta\mathbb{E}[\mathbb{E}[Y \mid \mathcal{A}]\mathbb{1}_A] = \mathbb{E}[(\alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$, and the bracket is $\mathcal{A}$-measurable.

(ii) $A := \{\mathbb{E}[X \mid \mathcal{A}] > \mathbb{E}[Y \mid \mathcal{A}]\} \in \mathcal{A}$ and $0 \ge \mathbb{E}[(X - Y)\mathbb{1}_A] = \mathbb{E}[(\mathbb{E}[X \mid \mathcal{A}] - \mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$ by (i); the integrand is > 0 on A, so $P(A) = 0$.

(iii) First X^+Y^+: take $\mathcal{A}$-elementary $Y_n^+ = \sum_i c_{i,n}\mathbb{1}_{A_{i,n}} \nearrow Y^+$. For $A \in \mathcal{A}$, the MCT twice gives

$$\mathbb{E}[X^+Y^+\mathbb{1}_A] = \lim_n \sum_i c_{i,n}\mathbb{E}[X^+\mathbb{1}_{A_{i,n}\cap A}] = \lim_n \sum_i c_{i,n}\mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]\mathbb{1}_{A_{i,n}\cap A}] = \mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]Y^+\mathbb{1}_A].$$

So $\mathbb{E}[X^+Y^+ \mid \mathcal{A}] = Y^+\mathbb{E}[X^+ \mid \mathcal{A}]$; the same holds for $X^\pm Y^\mp$, X^-Y^-, and linearity gives the claim.

(iv) $\mathbb{E}[X \mid \mathcal{A}_1]$ is $\mathcal{A}_2$-measurable, so the first expression equals $\mathbb{E}[X \mid \mathcal{A}_1]$ by Example 2.34(i). For $A \in \mathcal{A}_1 \subset \mathcal{A}_2$: $\mathbb{E}[\mathbb{1}_A\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1]] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_2]] = \mathbb{E}[\mathbb{1}_A X] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_1]]$.

(v) $\pm X \le |X|$ with (i), (ii).

(vi) $X \in \mathcal{L}^1$ (DCT); by (v), (i), $\mathbb{E}|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}|X_n - X| \to 0$. [Supplement] *A.s.:* $Z_n := \sup_{k\ge n}|X_k - X| \le 2Y$ decreases to 0, $|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[Z_n \mid \mathcal{A}] \searrow Z \ge 0$ (by (ii)), and $\mathbb{E}[Z] \le \mathbb{E}[Z_n] \to 0$ (DCT).

(vii) [Supplement] $\varphi = \sup_k \ell_k$ for countably many affine $\ell_k(x) = a_k x + b_k$ (supporting lines at rationals); by linearity and monotonicity, $\ell_k(\mathbb{E}[X \mid \mathcal{A}]) = \mathbb{E}[\ell_k(X) \mid \mathcal{A}] \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$ off a null N_k; take $\sup_k$ off $\bigcup_k N_k$. ■

Definition 2.36 (Conditional expectation of nonnegative r.v.'s)

For $\mathcal{F}$-measurable $X : \Omega \to [0, \infty]$, $\mathbb{E}[X \mid \mathcal{A}]$ is any Y with (i) Y $\mathcal{A}$-measurable and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is a.s. unique and a.s. $[0, \infty]$-valued.

Existence. $\mathbb{E}[X \wedge n \mid \mathcal{A}]$ increases a.s. (Theorem 2.35(ii)); $Y := \lim_n \max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]$ ($X_k := X \wedge k$) is $\mathcal{A}$-measurable and, by the MCT, for $A \in \mathcal{A}$

$$\mathbb{E}[X\mathbb{1}_A] = \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{A}]\mathbb{1}_A] = \lim_n \mathbb{E}[\max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A].$$

Uniqueness and positivity [Supplement]. For rational $a < b$, $A := \{Y' \le a < b \le Y\}$ has $bP(A) \le \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[Y'\mathbb{1}_A] \le aP(A) < \infty$, so $P(A) = 0$; hence $Y = Y'$ a.s. $A_n := \{Y < -\frac{1}{n}\}$ has $0 \le \mathbb{E}[Y\mathbb{1}_{A_n}] \le -\frac{1}{n}P(A_n)$, so $Y \ge 0$ a.s.

Likewise, for $X, Y \ge 0$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$; $X \in \mathcal{A} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$; $X = Y$ a.s. $\Rightarrow$ equal conditional expectations; linearity with coefficients $\alpha, \beta \ge 0$; and $X \le Y$ a.s. $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

Theorem 2.37

Let $X_n : \Omega \to [0, \infty]$ be $\mathcal{F}$-measurable; (in)equalities hold a.s.

(1) (Beppo Levi) $X_n \nearrow \Rightarrow \mathbb{E}[\sup_n X_n \mid \mathcal{A}] = \sup_n \mathbb{E}[X_n \mid \mathcal{A}]$.

(2) (Fatou) $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] \leq \liminf_n \mathbb{E}[X_n \mid \mathcal{A}]$.

Proof [Supplement]. (1) $Y_n := \mathbb{E}[X_n \mid \mathcal{A}] \nearrow Y := \sup_n Y_n$ a.s.; for $A \in \mathcal{A}$, the MCT twice: $\mathbb{E}[\sup_n X_n \mathbb{1}_A] = \lim_n \mathbb{E}[X_n \mathbb{1}_A] = \lim_n \mathbb{E}[Y_n \mathbb{1}_A] = \mathbb{E}[Y \mathbb{1}_A]$, so $Y = \mathbb{E}[\sup_n X_n \mid \mathcal{A}]$.

(2) $g_n := \inf_{k \geq n} X_k \nearrow \liminf_n X_n$, $\mathbb{E}[g_n \mid \mathcal{A}] \leq \mathbb{E}[X_k \mid \mathcal{A}]$ $(k \geq n)$; by (1) $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] = \lim_n \mathbb{E}[g_n \mid \mathcal{A}] \leq \lim_n \inf_{k \geq n} \mathbb{E}[X_k \mid \mathcal{A}]$. ■